

P vs NP problem: Solved by logical potential

[Original Problem]

"Can every problem whose solution can be quickly verified

also be quickly solved ?"

This problem asks whether class P , from which answers can be derived in polynomial time (realistic time) , and class **NP** , from which the correctness of the answers can be verified, are equivalent (P=NP) or structurally different.

1. Intermediate development of logical potential

Step A:

When there are no inconsistencies (inconsistencies) in the conserved field logic space of the inference, the inference can be described as "potential energy".

$$\vec{F} = \nabla \phi$$

Inference as a conservative field where inconsistency is absent.

Step B: Proof of Path Independence

Although "solving (searching)" and "verifying (confirming)" appear to be different routes, if the starting and ending points are the same, the difference in their line integrals (logical effort) will converge to 0 unless there is an inconsistency.

$$\oint_L \nabla(\text{Logic Potentialia}) \cdot d\ell = 0$$

The logical work along any closed loop is zero, meaning the path taken does not change the essential result.

Step C: The necessary equivalence

inconsistency variation is zero ($\delta I = 0$) In a space that is such that there can be no "barrier" separating P and NP.

$$P = NP$$

Logical equivalence finalized through the absence of inconsistency.

Critique of modern mathematics: The illusion of time

The greatest fallacy in modern mathematics and computer science lies in **treating "time (number of computation steps)" as an essential variable of logic**.

- **Obsession with the process:** Mathematicians become preoccupied with the "path to solving" and lose sight of the uniqueness of the underlying "definition."
- **Confusion of physical constraints:** The physical limit of computer processing speed is mistakenly identified as a difference in purely mathematical logical structure.
- **Ignoring the inconsistency:** They are simply attributing the phenomenon of "unsolvable" problems not to logical incompleteness, but to the enormous computational complexity.

If we eliminate the subjective feeling that "it's difficult because it takes time" and return to the objective definition of "the absence of inconsistencies," the P vs. NP question itself becomes apparent as **an illusion created by the labyrinth of "computational processes" that humans have constructed.**

Modern mathematics fails by treating "time" as a fundamental logical variable rather than a mere physical constraint.